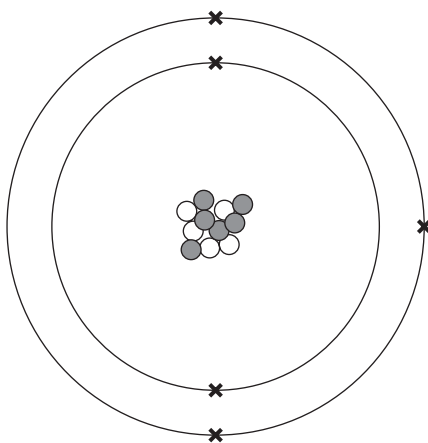


2. (a) Atoms contain particles called electrons, protons and neutrons. The diagram shows an atom of boron.



Complete the sentences.

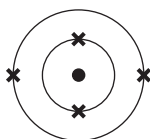
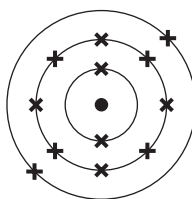
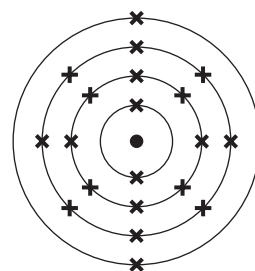
[4]

The nucleus of a boron atom contains five and
six

There are five in the shells around the nucleus.

The particles in the atom which have a positive charge are called
.....

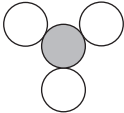
- (b) The diagrams show three different atoms labelled **A**, **B** and **C**.
These letters are **not** chemical symbols.

**A****B****C**

Complete the sentences.

- (i) Atoms **A**, **B** and **C** can be found in Group [1]
(ii) The atomic number of atom **A** is [1]
(iii) The electronic structure of atom **C** is (2,). [1]
(iv) Atom **B** can be found in Period [1]

- (c) The following table shows three substances, their formulae and diagrams that can be used to represent them.

Substance	Formula	Diagram
nitrogen trioxide	NO_3	
methane	CH_4	
water	H_2O	

- (i) Use the information in the table to work out the key being used to represent the different elements in the diagrams. [2]

 is

 is

 is

 is

- (ii) **Using the same key** draw a diagram to represent a molecule of carbon dioxide, CO_2 . [1]

- (d) The chemical formula of phosphoric acid is H_3PO_4 .

(i) State how many phosphorus atoms are present in the formula H_3PO_4 [1]

(ii) Give the **total** number of atoms shown in the formula H_3PO_4 [1]